

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Efficient posterior approximation is a recurring need across Bayesian inference problems, from diagnostic examples to inverse problems and predictive models. Stein Variational Gradient Descent (SVGD) is a particle-based variational inference method that transports an ensemble toward a target distribution, with its dynamics determined by a reproducing kernel Hilbert space (RKHS). Because the kernel controls particle interactions, the same SVGD update can behave differently across target geometries, dimensions, and posterior scales. The median heuristic gives a convenient bandwidth default, but its scalar and non-adaptive form can be poorly matched to anisotropic or higher-dimensional targets. We study kernel selection as a general mechanism for adapting SVGD within a prescribed family of kernels, rather than as a problem-specific tuning step. Motivated by convergence analyses based on the kernelized Stein discrepancy (KSD), we tune kernel parameters by maximizing the KSD during inference. The resulting method, Adaptive SVGD (Ad-SVG), alternates between the standard SVGD particle transport and gradient-ascent updates of kernel bandwidths. We provide a simplified theoretical analysis that lifts fixed-kernel KSD descent results to this adaptive setting and establishes convergence properties for the maximal KSD over the chosen kernel class. Across the numerical tasks considered here, Ad-SVG improves on standard bandwidth heuristics while preserving the underlying SVGD particle update.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based method for approximate Bayesian inference that complements traditional Markov Chain Monte Carlo (MCMC) methods. It evolves particles through update directions obtained from the functional gradient of the Kullback-Leibler (KL) divergence to a target distribution, with admissible directions constrained by the unit ball of a reproducing kernel Hilbert space (RKHS). Across this range of uses, the kernel is not merely an implementation detail: it shapes particle interactions and can therefore determine the practical performance of the algorithm [8, 28]. SVGD particle systems are also known to underestimate posterior variance in important regimes [2]. These limitations are often discussed as signs that SVGD becomes unreliable as dimension, anisotropy, or posterior scaling become more challenging. We revisit this issue from a focused but widely applicable angle: by dynamically tuning kernel parameters through maximization of the kernelized Stein discrepancy (KSD), we ask how much of SVGD’s behavior can be improved while keeping the basic particle-transport mechanism unchanged.

1.1 Related work.

Since its introduction [18], SVGD has been used as a general-purpose approximate inference tool in several machine learning settings [22, 24, 30, 15]. This breadth of use has motivated a parallel the-

oretical literature clarifying when and how the particle dynamics approach the target. Mean-field convergence has been studied in both continuous-time [23, 8, 5] and discrete-time [16, 32] formulations, while finite-particle convergence rates have been established under various assumptions [3, 33]. Other analyses interpret SVGD as a gradient flow on spaces of probability distributions [19, 8], in analogy with the variational formulation of diffusion processes in Jordan, Kinderlehrer, and Otto [14].

The performance of SVGD depends critically on the choice of the kernel function, as it determines the interaction between particles and the overall convergence of the method (see also Figures 1 and 9). Convergence results typically refer to mean-field convergence with respect to KSD, whose relation to weak convergence depends on the selected kernel [11]. The commonly used median heuristic [12] supplies a simple bandwidth rule, but it has limited theoretical support and can deteriorate as dimension, scaling, or anisotropy make a single global bandwidth less representative [2, 40, 36]. Several recent SVGD variants address related degradation mechanisms through more structured updates or projections [7, 37, 10, 21]. Other work expands the update class by using neural networks to learn transport directions [17, 39].

The closest methodological connection is Ai et al. [1], which studies a mixture-of-kernels framework. Their method defines a KSD for a weighted linear combination of kernels and learns the weights by maximizing this multiple-kernel KSD. Our formulation addresses a complementary scope: instead of restricting adaptation to a finite kernel basis, it optimizes continuous kernel parameters inside the same family that defines the SVGD update.

1.2 Contributions.

This work asks how broadly the original SVGD mechanism can remain effective when the kernel is selected adaptively rather than fixed by a default heuristic. By isolating kernel selection from architectural changes, neural parametrizations, and alternative transport rules, we study a scope of improvements that can be attached to the base method while preserving its familiar particle update. This framing lets us assess, within the experiments and assumptions considered here, how much of SVGD’s empirical behavior is governed by the kernel family and its parameterization.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce a kernel-selection mechanism that updates kernel parameters during SVGD inference by maximizing the KSD. Compared with the median heuristic, which fixes a scalar bandwidth rule, the method accommodates multiple continuous parameters and therefore applies to richer parameterized kernel families within the same SVGD framework.
- (ii) **Theoretical Analysis.** We analyze the algorithm in the discrete-time mean-field setting by extending fixed-kernel convergence results for SVGD. In this setting, the supremum of the KSD over a parameterized kernel class converges to zero, and under a Stein logarithmic Sobolev inequality we obtain iteration-complexity bounds for the mean-field limit.
- (iii) **Empirical Validation.** Across the numerical examples in the paper, adaptive kernel selection improves over the median heuristic and reduces the variance-collapse behavior observed for standard bandwidth choices.

2 Mathematical background

We introduce notation for the finite-dimensional inference settings used in the analysis and experiments. Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a symmetric positive definite kernel with associated RKHS $\mathcal{H}_0$. We define $\mathcal{H}$ as the d -fold Cartesian product of $\mathcal{H}_0$ with inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ is the subspace with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ denotes the set of couplings of μ and ν , namely probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] compare probability measures by maximizing integration discrepancies over a chosen test-function class. For Bayesian targets available only up to normalization, Stein’s method [35] makes this comparison usable by constructing target-centered test functions. In particular, for $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ holds for all $f \in \mathcal{H}$ [6, 20]. This leads to the *kernelized Stein discrepancy* (KSD), defined as

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The optimizer has the form $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map of k . Thus, for any fixed kernel satisfying the required conditions, the supremum reduces to the RKHS norm of ψ , yielding $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial reference distribution μ_0 , SVGD constructs a sequence of transported measures intended to approximate π . At iteration n , it sets $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp} \cdot$ is the push-forward under $T(x) = x + \gamma \psi^{\mu_n}(x)$ and $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the steepest-descent direction induced by the chosen RKHS. This update is motivated by the solution of Equation (1): $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes the KL-divergence. For this direction,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \middle| \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these maps repeatedly to particles $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the standard finite-particle update:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence for SVGD is difficult to prove without strong assumptions, several analyses use the kernelized Stein discrepancy (KSD) as a more tractable convergence quantity [16, 33, 32]. This perspective also makes the scope of kernel choice explicit. Minimizing a KSD for one fixed RKHS does not by itself guarantee weak convergence, since the mean-field measures may fail to be tight, a necessary condition for weak convergence on Polish spaces. As Equation (1) shows, the RKHS induced by the kernel determines which discrepancies are detected and therefore how KSD convergence can relate to weak convergence [11]. Consequently, adapting the kernel can be viewed as adapting the discrepancy used to guide SVGD, not only as tuning a numerical bandwidth.

We therefore formulate the adaptive variant over a parameterized family of kernels $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, KSD_θ denotes the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the optimal update direction, and Φ_θ the associated feature map. This parameterized view keeps the SVGD transport rule fixed while allowing the RKHS itself to respond to the current particle distribution. The resulting scope is broader than selecting a single bandwidth once, but remains bounded by the user-specified kernel class.

The selection rule follows the principle of choosing a kernel that maximizes the KSD between the current particle distribution and the target. The motivation is local but applies to any kernel family for which the KSD can be optimized: the instantaneous KL decrease under SVGD is proportional to the squared KSD at the current measure (see Equations (2) and (7)). Thus, within the available family, selecting a larger-KSD kernel seeks the steepest available KL decrease at that iteration. Geometric analyses of SVGD as a gradient flow support this interpretation by relating larger KSD values to more favorable convergence behavior when comparing the corresponding RKHSs [28, 8].

Standard SVGD typically studies convergence for one fixed kernel, whereas Ad-SVGd targets the maximal KSD over the selected kernel class. Relative to the median heuristic of Liu and Wang [18], this formulation is not tied to a specific scalar bandwidth rule: at each step, the idealized objective is

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

In the implemented algorithm, we approximate this objective by taking one or more gradient-ascent steps with step size $s > 0$ on the KSD before the particle update. The parameter update can also be skipped on selected SVGD iterations to reduce runtime (see Algorithm 1), controlled by the user-specified decision variable `paramupdate`. This separation keeps the kernel-selection mechanism compatible with other SVGD transport variants, such as adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates, and also permits replacing gradient ascent by another optimizer for the kernel parameters. The concrete implementation evaluated in this paper uses the representation [6, 20]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k \, d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the kernel-parameter gradient can be computed directly. The additional cost relative to vanilla SVGD is controlled in two generally useful ways:

- Kernel-parameter ascent reuses the same gradients $\nabla \log \pi(X_n^i)$ needed by the corresponding SVGD step, so Ad-SVGd does not require extra gradient evaluations of $\log \pi$.
- The update frequency and the number of kernel steps `nstepstheta` can be reduced, and for large particle sets the KSD objective can be approximated using subsampled particles.

Algorithm 1: Ad-SVGd

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps `nsteps`, `nstepstheta` $\in \mathbb{N}$
Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to nsteps  $- 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to nstepstheta  $- 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We analyze target measures with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$, a setting that covers the finite-dimensional targets used in our experiments and many standard Bayesian models after discretization. The purpose of the analysis is to show that fixed-kernel KSD convergence theory can be lifted to a kernel class when the relevant assumptions hold uniformly over parameters. Concretely, following [32], we adapt the analysis of [32] under Talagrand’s inequality, which is weaker than the logarithmic Sobolev inequality commonly used in related convergence arguments.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality $T1$, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

We first recall the fixed-kernel descent mechanism that will be reused for the adaptive family. Denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and use the following inequality as the common descent ingredient. For a fixed kernel parameter $\theta \in \Theta$ satisfying Assumption 3, define

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ satisfy $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Proposition 3.1 of [32] then gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. Applying this bound to the SVGD iterates (3) with a fixed $\theta \in \Theta$ yields the descent condition [32, Theorem 3.2]

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The crucial verification for (7) is $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, obtained from Talagrand's inequality in Assumption 2; this permits using (6) with $g = \psi_\theta^{\mu_n}$. The descent condition (7) then implies

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$.

With an iteration-dependent kernel parameter (θ_n) , the same fixed-kernel inequality (7) becomes

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

We next consider the ideal adaptive case in which the kernel parameter maximization is solved exactly at each step:

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

so that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi).$$

This exact formulation is available when the KSD maximization over the kernel parameter has a closed-form solution. One example is the multiple-kernel SVGD framework [1], where the kernel is a convex combination of base kernels

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For continuous or otherwise more general kernel families, the update θ_n in (9) typically has to be approximated. Throughout this idealized statement we assume that $\arg\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

Uniformity of the fixed-kernel assumptions over the class gives the following immediate descent statement from Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The practical algorithm only approximately maximizes the KSD over kernel parameters, so we also analyze an abstract alternating scheme. Suppose that $\max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n \end{aligned} \tag{10}$$

where $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ denotes any iterative rule intended to increase $\text{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is implemented by gradient ascent on the KSD, but the analysis is stated for a broader class of optimizers satisfying the following approximation property.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this approximation condition, formalized in Assumption 4, the adaptive scheme inherits convergence of the maximal KSD over the whole kernel class.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under a Stein logarithmic Sobolev inequality, this maximal-KSD statement can be translated into convergence in KL divergence. For several regimes of the approximation error ε_n , Section B derives iteration-complexity bounds for the mean-field system (10) and describes the bias that remains when KSD maximization is solved only to fixed $\bar{\varepsilon}$ -accuracy. Importantly, the condition is formulated over the kernel class, so it is enough for one admissible kernel in $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality.

5 Numerical Experiments

We evaluate Ad-SVGd on examples that probe complementary aspects of kernel choice: a one-dimensional Gaussian mixture, a linear inverse problem governed by an ordinary differential equation, and Bayesian logistic regression on a benchmark data set in Section 5.4. Additional experiments in Section A examine Gaussian mixtures, increasing dimension for multivariate normal targets, and Gaussian process inference. These studies keep the datasets, targets, and SVGd update conditions fixed while varying the kernel-selection strategy, so the results isolate how adaptive bandwidths transfer across different inference scenarios rather than introducing new model classes or evaluation protocols.

5.1 Kernel parameterization

A standard SVGd implementation uses kernels of the form

$$k_h(x, y) = \exp \left(-\frac{\|x - y\|_p^p}{h} \right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. We focus on bandwidth selection because this scalar controls the strength and range of particle interactions in many SVGd deployments. The median

heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the current median inter-particle distance, with the goal of balancing the attraction and repulsion terms in (3) [18].

To use the adaptive mechanism in settings where different coordinates may have different posterior scales, we choose product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with $h = (h_1, \dots, h_d)$, allowing a separate bandwidth for each coordinate. The derivatives required by Algorithm 1 follow directly from Equations (4) and (5). We also tested a coordinate-wise version of the median heuristic for these kernels, using a naive median in each dimension; in the settings considered here, that alternative did not perform well and exhibited variance collapse.

The same adaptation principle can be applied to other parameterized kernel families beyond the product kernels used in the experiments. For higher-dimensional problems, spectral or low-rank parametrizations could target dominant variability directions without the full cost of coordinate-wise products, while Matérn or mixed-product constructions could vary smoothness and anisotropy. We do not evaluate these alternatives here, but they illustrate that the method is defined at the level of a kernel class rather than a particular bandwidth formula.

5.2 Toy example

We first use the one-dimensional Gaussian mixture example from Liu and Wang [18], with target $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. This diagnostic example keeps the target simple while exposing how sensitive the particle approximation can be to bandwidth scale. We run SVGD with $M \in \{50, 200, 500\}$ particles initialized from $\mathcal{N}(0, 1)$ for 10^4 steps with step size 1, using the kernels above with $p = 1$ and fixed choices of bandwidth h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit one-dimensional formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) using an exact sample of size 10^5 . As shown in Figure 1, strong performance is confined to a limited bandwidth range. Even in this low-dimensional diagnostic case, the result illustrates a more general issue: fixed bandwidths encode assumptions about particle scale that may not remain appropriate across targets or particle counts.

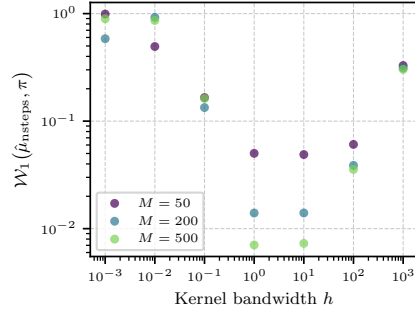


Figure 1: Final Wasserstein 1-distances in the one-dimensional Gaussian mixture example, illustrating the dependence of fixed-kernel SVGD on the bandwidth h .

5.3 Linear inverse problem based on ODE

The next experiment uses a discretized inverse problem adapted from [38, Example 2.1]. It represents a common Bayesian inverse-problem setting in which the posterior is finite-dimensional after truncation but inherits anisotropy from an underlying function-space model. We consider the

one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from noisy point observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise. The forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ evaluating the solution at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, we place a Gaussian process (GP) prior on u through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients are independent with $x_k \sim \mathcal{N}(0, \lambda_k)$ and $\lambda_k = 50k^{-2}$. After truncation, inference reduces to estimating $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under prior $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. With Gaussian noise $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density is

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation, H is replaced by a numerical discretization of Equation (11) on a grid with mesh size 2^{-8} , and we use the fully observed system with $N_{\text{obs}} = 2^8$. The KL expansion uses $N_x = 16$ terms, the noise covariance is $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$, and reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD with the median heuristic (Med-SVGD) and Ad-SVGD for particle ensemble sizes $M \in \{50, 100, 200\}$. Both methods use the kernels from Section 5.1 with $p = 1$ and run for $4 \cdot 10^5$ iterations with particle step size 10^{-3} and the AdaGrad-style adaptive step-size control suggested by Liu and Wang [18]. For Ad-SVGD, bandwidth updates use step size 10^{-5} and are performed once every 100 particle updates; under this schedule, the adaptive rule adds no significant runtime difference in our implementation.

Figure 2 shows one representative GP reconstruction from this discretized inverse problem. Both methods approximate the posterior mean, but Ad-SVGD better captures posterior uncertainty. To quantify this behavior across seeds, we compute the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian approximation induced by the particle mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [29, Section 3.2]. Since π is multivariate normal, the distance has the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

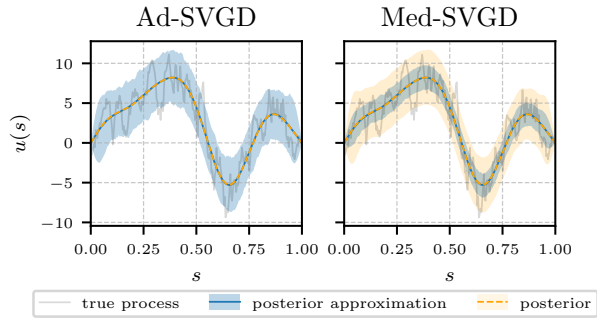
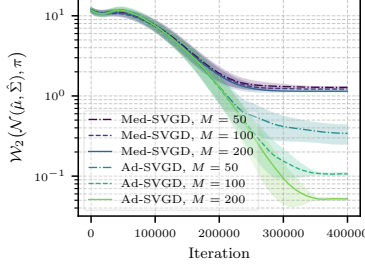
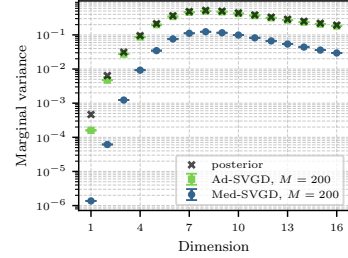


Figure 2: GP reconstruction in the ODE-based inverse problem, showing mean and 90% confidence interval.

where μ_π and Σ_π are the posterior mean and covariance. We also compare marginal variances of the final particle distribution with those of the posterior. Figure 3 aggregates the results over 56 random seeds; the posterior covariance is independent of $\bar{y}$. In this inverse-problem setting, Ad-SVGD gives closer posterior approximations than Med-SVGD, which underestimates uncertainty. Unlike Med-SVGD, Ad-SVGD also benefits from increasing the particle count beyond 50. Figure 4 shows that the learned component-wise bandwidths stabilize earlier than the approximation error, differ substantially across coordinates, and are negatively correlated with the corresponding marginal variances, consistent with the broader goal of adapting interactions to posterior scale.

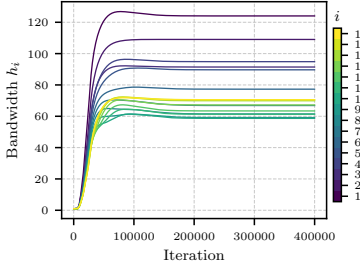


(a) Approximation-error convergence

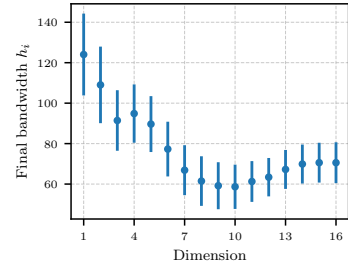


(b) Posterior and particle marginal variances

Figure 3: Aggregated ODE inverse-problem results comparing Med-SVGD and Ad-SVGD across particle counts, reported as means and 95% confidence intervals over 56 random seeds.



(a) Mean bandwidth evolution by dimension



(b) Final bandwidths with 95% confidence intervals

Figure 4: Adaptive bandwidth behavior in the ODE inverse problem, showing how Ad-SVGD learns coordinate-dependent interaction scales over 56 random seeds.

5.4 Bayesian logistic regression

To test the method in a supervised Bayesian model used in prior SVGD work, we follow Liu and Wang [18] and Liu et al. [21] and consider Bayesian logistic regression (BLR) on the Covertype data set [4]. This benchmark complements the synthetic and inverse-problem examples by evaluating the same adaptive-kernel rule in a data-driven posterior approximation task. The regression weights w have prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and the inference target is the posterior of $x = [w, \alpha]$. As in Liu et al. [21], we use the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] to construct reference samples for evaluating Med-SVGD and Ad-SVGD. Following Liu et al. [21],

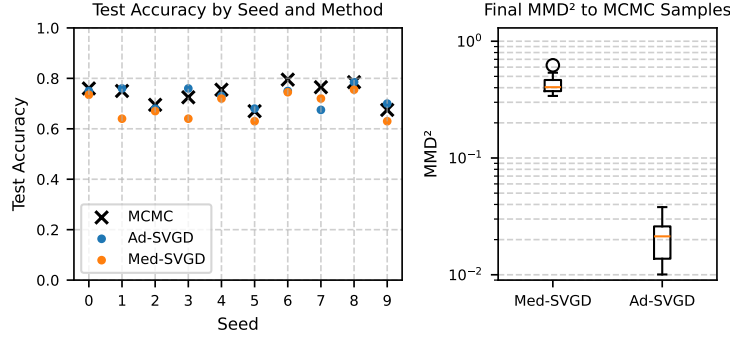


Figure 5: Bayesian logistic regression approximation quality on Covertypes, measured by prediction accuracy (left) and MMD^2 to NUTS reference samples (right).

we evaluate on subsets of size 1000 from the original data set and aggregate over 10 random draws. Classification performance is measured by test-set prediction accuracy using the particle mean. As shown in Figure 5, Ad-SVGD improves over Med-SVGD for almost all seeds, with higher test accuracy and closer agreement with the MCMC reference under the same data splits and model specification.

The same figure also reports squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples, providing a posterior-approximation metric beyond prediction accuracy. The gap between the methods is also visible in Figure 7, which compares final-particle covariance matrices with the MCMC reference, averaged over 10 seeds using the visualization style of Liu et al. [21]. Med-SVGD substantially underestimates posterior uncertainty, whereas Ad-SVGD is closer to the reference covariance. Figure 6 further shows that, starting from the uninformed value 1 in every component, Ad-SVGD learns component-wise bandwidths that differ by several orders of magnitude, indicating that the adaptive rule is not restricted to the synthetic geometries used earlier.

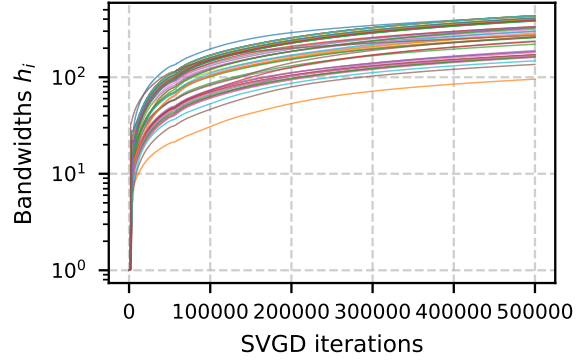


Figure 6: Adaptive bandwidth evolution in the BLR example, showing component-wise scales learned during posterior approximation.

6 Limitations and outlook

The main limitation of the theory is its reliance on Assumption 4, which abstracts the quality of the kernel-parameter optimization. Our experiments use the alternating gradient-ascent scheme in Algorithm 1, but the paper does not prove that this concrete update always satisfies Assumption 4. The empirical results therefore support the usefulness of adaptive kernel selection under the tested conditions, while the precise optimization conditions required for the assumption remain a subject

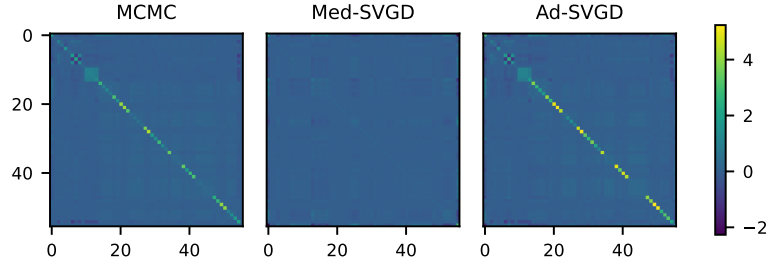


Figure 7: Covariance matrices of final SVGD particle sets compared with MCMC samples in BLR, averaged over 10 seeds.

for further work across broader kernel families and target classes.

Our analysis and experiments deliberately focus on the original SVGD dynamics in order to isolate the effect of kernel adaptation. A natural next step is to combine the same kernel-selection principle with broader SVGD variants, including sliced SVGD [10], Grassmann SVGD [21], and Stein transport [27]. Such extensions would test whether KSD-driven kernel selection remains useful when the transport map, projection structure, or geometry of the update is also changed.

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A More numerical experiments

A.1 Gaussian mixture models

We first revisit the diagnostic mixture from Section 5.2. For the setup described there, Figure 9 shows how fixed bandwidth choices $h \in \{0.001, 1, 1000\}$ alter the final particle distribution relative to the target density. We then compare Med-SVGD and Ad-SVGD across particle counts M , using the final particle distributions after 10^4 iterations with step size 1. This appendix experiment clarifies the low-dimensional regime in which a simple scalar bandwidth can already be adequate.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle distribution and the target for Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. Approximation quality improves with particle count for both methods, and in this simple one-dimensional setting both reach a Wasserstein distance below 0.01 for $N = 500$. Thus the advantage of adaptation is not claimed to be universal; it is most relevant when the kernel family needs to represent nontrivial scale variation.

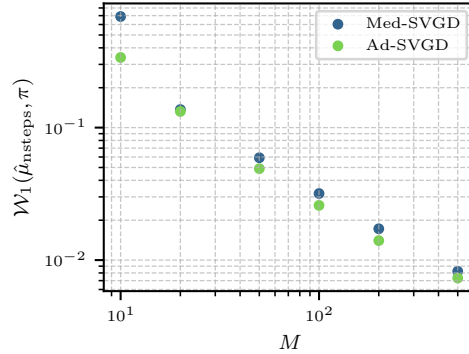


Figure 8: Approximation error for the one-dimensional Gaussian mixture across particle counts.

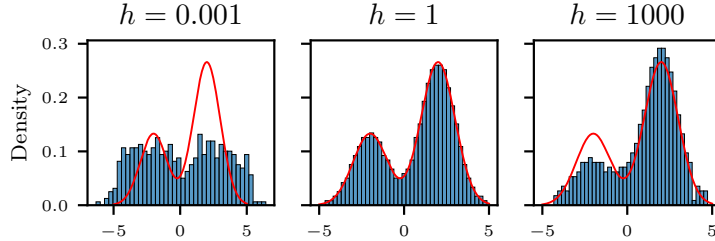


Figure 9: Final particle histograms for fixed bandwidths $h = 0.001, 1, 1000$ in the one-dimensional mixture, showing how bandwidth scale changes the approximation; the target density π is shown in red.

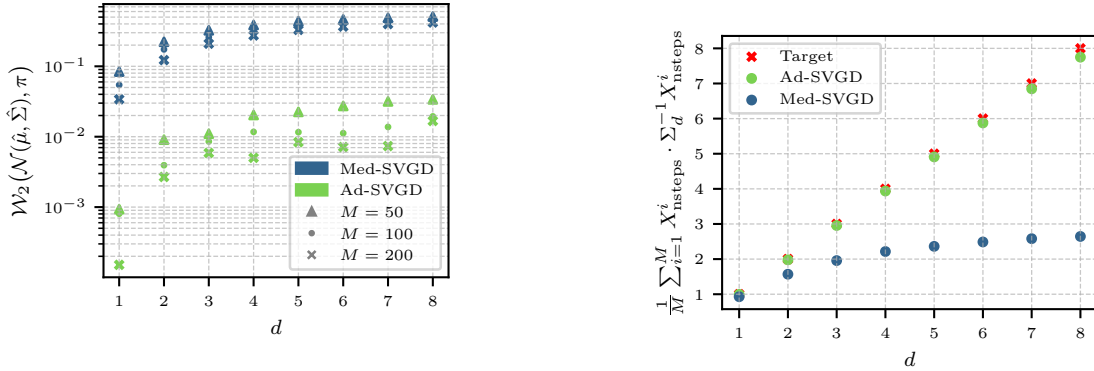
In this one-dimensional case, both adaptive and median-based bandwidth selection approximate the target well. The result is useful as a boundary case: Ad-SVGD remains competitive, but the extra flexibility does not provide a clear advantage when a scalar bandwidth is already sufficient for the target geometry.

A.2 Scaling dimension: multivariate normal distribution

To isolate the effect of anisotropy as dimension grows, we next consider Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The initial particle distribution is

$\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and the algorithms are run for 10^4 iterations with step size 0.1. When needed for numerical stability, we use a smaller step size and adjust the number of iterations accordingly.

We compare the final particles from Med-SVGD and Ad-SVGD as dimension increases. This controlled Gaussian family isolates dimensional scaling and anisotropic variance from other modeling complications. Because the Wasserstein 2-distance between Gaussian distributions has an explicit formula (see [29]), sample quality is measured by the Wasserstein 2-distance between π_d and $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance of the particles [29, Section 3.2]. Following the naming of the Python Optimal Transport implementation used for our calculations [9], we refer to this quantity as the *Bures Wasserstein distance*. Figure 10a shows this metric for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as d increases; Ad-SVGD is consistently better across the tested dimensions.



(a) Approximation error across dimensions for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic across dimensions for $M = 200$.

Figure 10: Results for anisotropic multivariate normal targets, isolating how bandwidth adaptation behaves as dimension increases.

The Bures Wasserstein distance is less sensitive to errors in the low-variance coordinates of π_d . To expose these errors, we also evaluate the statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom and has expectation d when $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We compute the mean statistic over particles with $M = 200$ and compare it with d in Figure 10b. As dimension increases, the Med-SVGD statistic moves progressively away from the target expectation, whereas the Ad-SVGD statistic remains comparatively close, indicating better preservation of uncertainty across coordinates in this controlled scaling study.

Tables 1 and 2 provide coordinate-level details for $M = 200$ and different d . Each table lists particle marginal variances against the target marginal variances in the first row; Table 1 reports Med-SVGD and Table 2 reports Ad-SVGD. The adaptive method tracks the target variances across components much more closely, while Med-SVGD underestimates uncertainty, consistent with Figure 10b.

Finally, for $M = 200$ and $d = 8$, we examine the normalized marginal particle distributions. Each coordinate is scaled by the inverse target marginal standard deviation, equivalently multiplying the i -th component by i , so the target marginals become standard normal. Figures 11 and 12 show the resulting histograms for Med-SVGD and Ad-SVGD with a standard Gaussian density for reference. Ad-SVGD captures the normalized marginals more accurately, whereas Med-SVGD remains too concentrated. The same pattern appears in Figures 13 and 14, which compare normalized marginal particle quantiles with standard normal quantiles.

Table 1: Marginal variances of final Med-SVGD particle distributions for the d -dimensional Gaussian examples with $M = 200$, compared with target variances for π_d .

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of final Ad-SVGD particle distributions for the d -dimensional Gaussian examples with $M = 200$, compared with target variances for π_d .

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

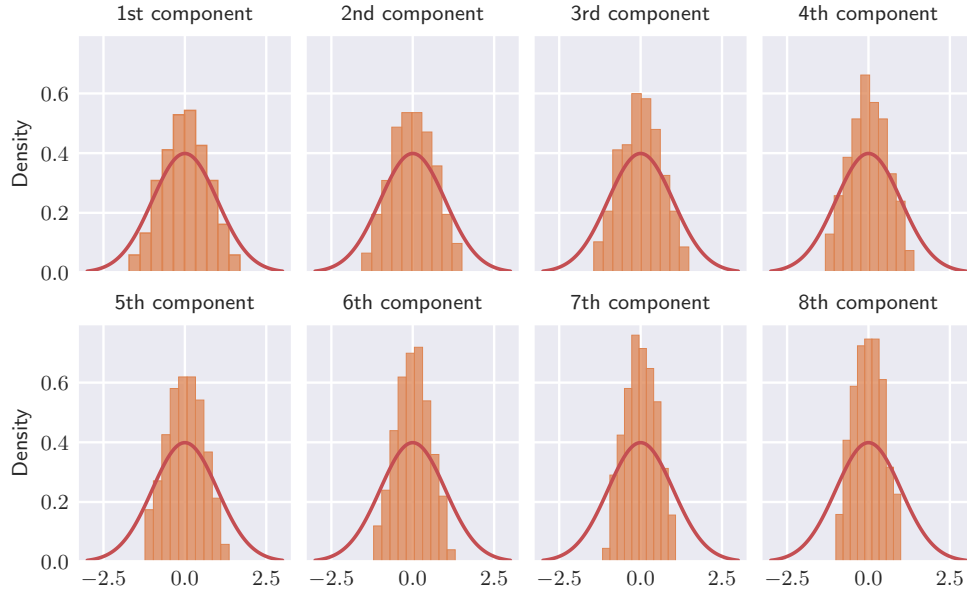


Figure 11: Coordinate-wise histograms of final Med-SVG particles for the eight-dimensional Gaussian example with $M = 200$, rescaled by marginal target variances to assess normalized marginal fit; the standard Gaussian density is shown for comparison.

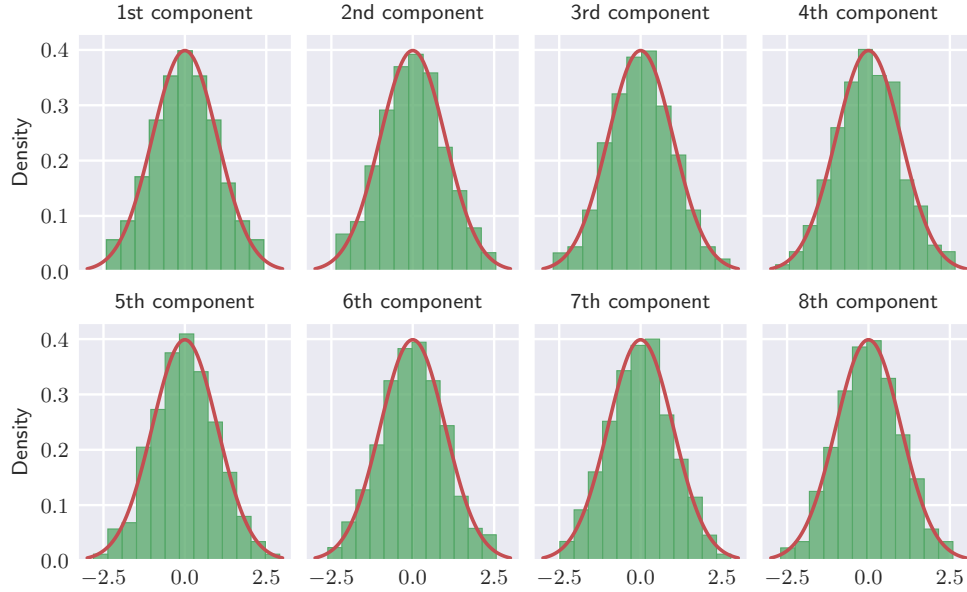


Figure 12: Coordinate-wise histograms of final Ad-SVG particles for the eight-dimensional Gaussian example with $M = 200$, rescaled by marginal target variances to assess normalized marginal fit; the standard Gaussian density is shown for comparison.

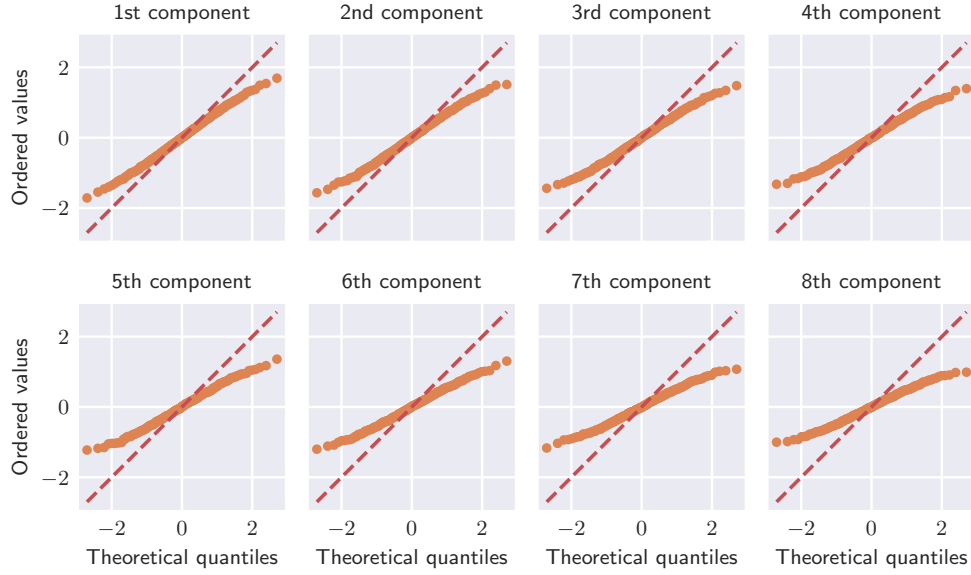


Figure 13: Q-Q plots comparing rescaled final Med-SVGD particle marginals for the eight-dimensional Gaussian example with a standard normal distribution, providing a coordinate-wise check of Gaussian fit; the reference line has slope 1 and passes through the origin.

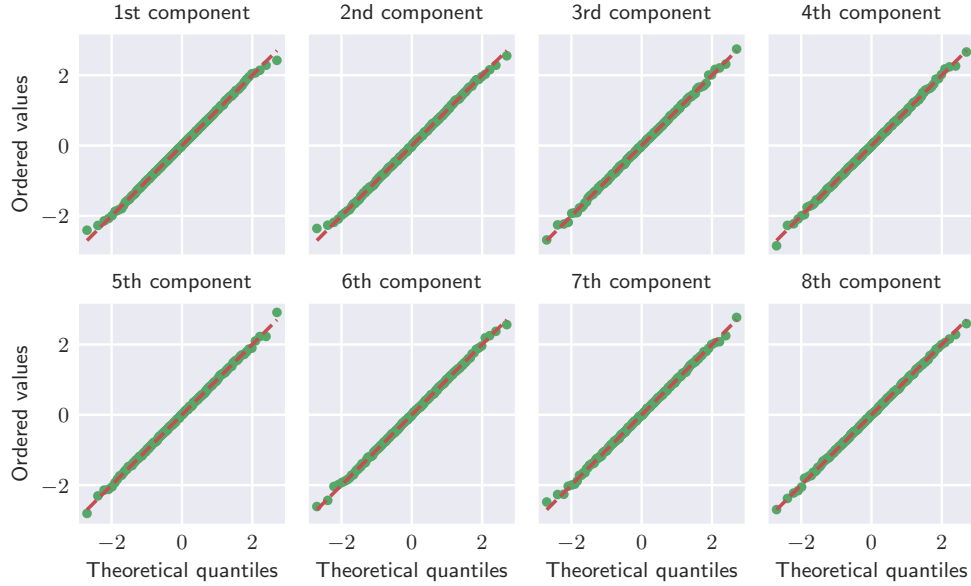


Figure 14: Q-Q plots comparing rescaled final Ad-SVGD particle marginals for the eight-dimensional Gaussian example with a standard normal distribution, providing a coordinate-wise check of Gaussian fit; the reference line has slope 1 and passes through the origin.

A.3 Gaussian process inference

We also test Ad-SVGD in a Gaussian process (GP) inference task from [31], which provides another finite-dimensional inverse problem with controllable truncation dimension. The GP on $[0, 1]$ is represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ has independent entries $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this gives the inverse problem $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ with $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal entries k^{-2} , and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$. This setup extends the inverse-problem evidence to a GP coefficient posterior while preserving a closed-form Gaussian reference.

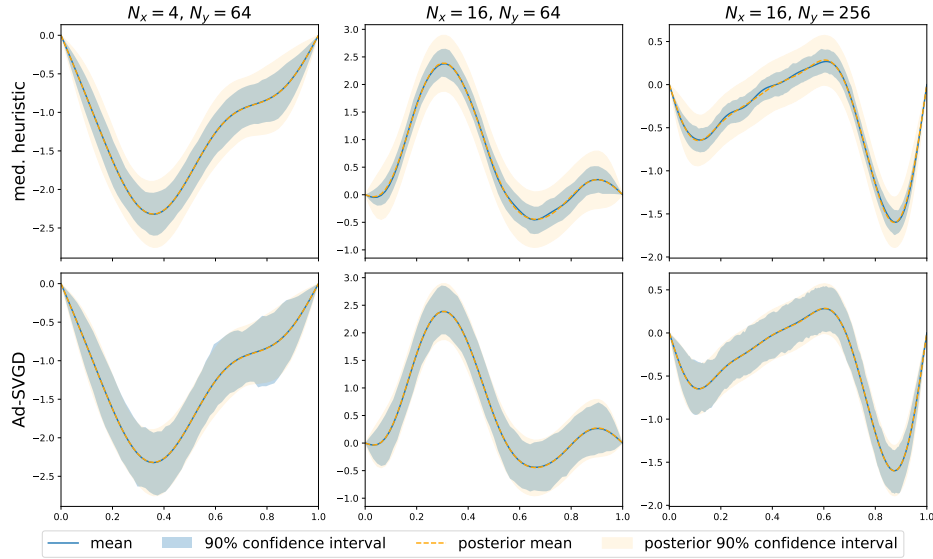


Figure 15: Estimated processes from the median heuristic and Ad-SVGD with $M = 100$ particles for different GP model configurations, compared with the posterior reference.

We use SVGD to approximate the posterior $X | Y = \bar{y}$ and compare the median heuristic with the adaptive kernel rule. Med-SVGD uses kernels $k(x, y) = \exp(-\|x - y\|_1/h)$, while Ad-SVGD uses product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with $h = (h_1, \dots, h_{N_x})$, assigning a bandwidth to each coefficient dimension. Following [18], the scenarios with $N_x = 16$ use an Adagrad variant for step-size control.

Figure 15 compares the resulting processes with the posterior for $N_x = 4$, $N_y = 64$; $N_x = 16$, $N_y = 64$; and $N_x = 16$, $N_y = 256$. As truncation dimension increases, the median heuristic underestimates posterior variance, while Ad-SVGD gives a closer approximation. This pattern persists across the tested numbers of GP observations. Table 3 quantifies the effect through the trace of the particle covariance compared with the theoretical posterior covariance: Med-SVGD substantially underestimates uncertainty, whereas Ad-SVGD is much closer to the posterior trace across the reported GP configurations.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the final particle covariance ($M = 100$ particles) compared with the theoretical posterior covariance across GP configurations, averaged over 25 runs.

B Refined theoretical analysis

This final section records stronger KL convergence guarantees under a Stein log-Sobolev inequality, which relates the KL divergence to the KSD. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it might hold. In the adaptive setting, the condition can be stated at the level of the kernel class: it is enough that the maximal KSD over the class controls the KL divergence, which may hold even if only part of the class provides the necessary control. This formulation is broader than requiring every fixed kernel in the family to provide the same KL control. More precisely, we assume the following.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

With this property, maximal-KSD control over the class yields KL convergence bounds. In particular, the argument also covers a fixed optimization-accuracy regime $\varepsilon_n \leq \bar{\varepsilon}$, so ε_n need not vanish in order to obtain an explicit bias bound for $\mathcal{F}(\mu_n)$. The result therefore separates the statistical effect of using a kernel class from the optimization accuracy with which the best kernel parameter is found.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}$$

for all $n \in \mathbb{N}$.

Proof. As in the proof of Theorem 3, Assumption 5 and the stated conditions on α, γ give

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$ yields the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

The three stated approximation regimes are handled separately:

1. If $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows from the geometric-series bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1-\rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1-\rho/q} = q^n \frac{1}{q-\rho}$$

using the formula for a geometric series. Similarly, when $q < \rho$, we directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1-q/\rho} = \rho^n \frac{1}{(\rho-q)}.$$

Finally, when $q = \rho$, bounding $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ gives the stated rate by the same argument.

3. For $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor+1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here we use $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□